

Unit 5

THE RULES OF THE GAME:WHO GETS
WHAT AND WHY

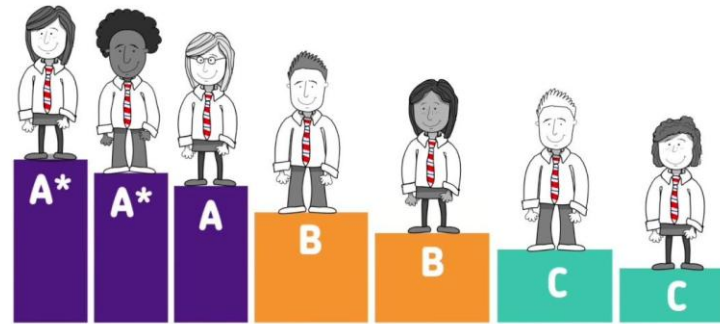
Who Gets What — and Why?

- Imagine you apply for a part-time job.
- Employer says: 'Take it or leave it.'

Is this deal FAIR? Is it EFFICIENT? Who decides the rules? ->

Institutions

Rules of the game = institutions



- rules shape *outcomes* and *choices*



Power

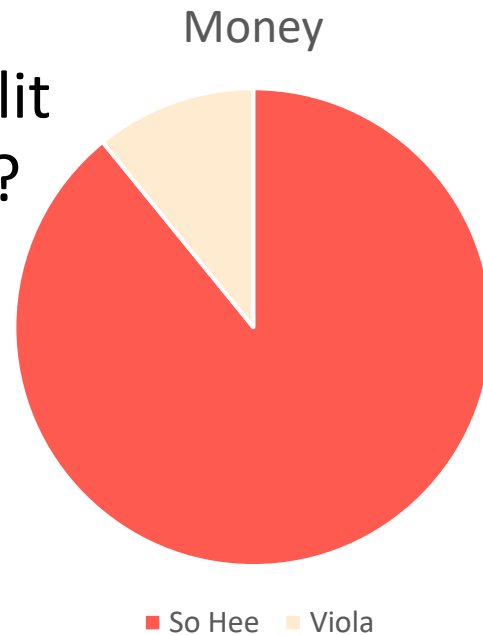
Power is the ability to do and get the things we want in opposition to the intentions of others.

Structural power = outside options (*what happens if you walk away*).

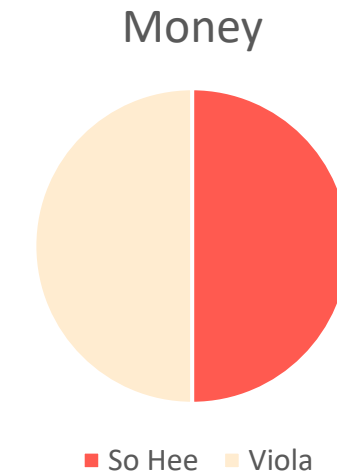
Bargaining power = ability to impose terms.

Efficiency vs Fairness

Big Pie → 90/10 split
Efficient but unfair?



Think: *which would you choose?*



Smaller Pie → 50/50 split
Fair but less efficient?

Efficiency (economic sense)

- An outcome is **efficient** (Pareto efficient) if **no one can be made better off without making someone else worse off**.
- It says nothing about who gets more or less — only that all resources are used optimally.
- Example:
 - If Angela produces 10 apples, and Eduardo takes 9 while Angela keeps 1 → **efficient** (no apples wasted).
 - But clearly **not fair**.

Fairness (equity sense)

- **Fair** means people see the outcome as *just* or *reasonable*.
- It depends on values, social norms, and rules.
- Example:
 - Angela and Eduardo split 10 apples equally (5–5). Many people would call this **fair**.
 - But if production was reduced to 8 apples in the process, it may be **less efficient** than the 9–1 split.

Example: Both Fair and Efficient

- Suppose Angela and Eduardo work together and produce **10 apples**.
- They split them **5–5**.
- Outcome:
 - **Efficient** → all apples are used, no waste.
 - **Fair** → equal division matches many people's sense of justice.

Putting it together

- **Efficiency** = Are we making the most of the pie?
- **Fairness** = How is the pie divided?
- Sometimes outcomes can be **efficient but unfair** (big pie, very unequal split).
- Sometimes outcomes can be **fair but inefficient** (equal split, but smaller pie).
- The challenge in economics is: can we design rules that achieve **both**?

Angela & Bruno: Rules of the Game

Scenario	Angela's outcome	Bruno's outcome	Rule type
Forced labour	Very low	Very high	No outside option
Voluntary contract	Moderate	High	Outside option improves bargaining

- Angela's wellbeing depends not only on productivity, but on the RULES of the game and her outside options.
 - If she has no choice, Bruno captures most gains.
 - If she can walk away, bargaining shifts.

OUTLINE

- A. Topic 1 - Introduction
- B. Topic 2 - Angela and Bruno: Three Cases
- C. Topic 3 - Measuring, understanding and addressing inequality

A. Topic 1 - Introduction

Institutions, rules of the game

- Common source of confusion is the term institutions.
- Institutions do not refer to the layman's definition of physical buildings or organizations.
- Rather, institutions are the “rules of the game”.

Institutions

- **Institutions** are the “rules of the game”.
- They govern the interactions between individuals.
- They provide the constraints and the **incentives**.
- They specify who can do what, when they can do it, and how the actions of players (in game theory terminology) affect their payoffs.

Ultimatum game vs Dictator game

- To *see* how institutions can affect outcomes, let's play both the ultimatum game and dictator game, then compare the results.
 - **Ultimatum** game, one student is given a sum of tokens and must propose how to divide it between themselves and other students. The other students can either accept or reject the offer. If the offer is accepted, both players receive the proposed amounts. If it is rejected, neither receives anything.
 - **Dictator** game, one student is again given a sum of tokens and decides how to divide it between themselves and other students. This time, however, the other students have no power to accept or reject. Whatever the first student decides is final.

Student clubs and societies

- What is the role of the president?
- How are they selected?
- Does anyone in the club have veto power over decisions?

Power

- **Power** is the ability to do and get the things we want in opposition to the intentions of others.
- Power in voluntary economic interactions takes two forms:
 - Structural power - When we consider conflicts over how something of value will be divided, a person's structural power is the value of their next best alternative. A person's structural power is limited by the structural power of the other.
 - Bargaining power - A person exercising bargaining power may set the terms of an exchange and/or impose, or threaten to impose, heavy costs.

Power

- *Structural* power has to do with one's own reservation option (next best alternative) and the ability to influence the other player's reservation option.
- Once the reservation options of the two players have determined the most and least that each player can attain (such that the interaction will take place), *bargaining* power determines how the surplus is divided between the two players.

Example: Structural vs Bargaining Power

- **Story**

- Shota wants to form a study group with Seowon
- Shota has **no alternative** if Seowon refuses.
- Seowon already has another group to fall back on.

- **Structural Power**

- Comes from outside options.
- Seowon's reservation option is strong, Shota's is weak.
- This gives Seowon more leverage before bargaining even starts.

- **Bargaining Power**

- Once they decide to study together, the total benefit (better grades, saved time) is the **surplus**.
- Bargaining power decides **how that surplus is split**.
- Seowon might push Shota to do more of the homework, or they might share fairly.

- **Takeaway**

- Structural power sets the limits. Bargaining power decides the split inside those limits.

Discussion: Structural vs Bargaining Power

- Can you think of people or groups who have a strong reservation option in a specific situation? Or a weak reservation option?
- For example,
 - immigrant workers whose residency depends on their employer may have a weak reservation option because if they lose the job they may have to leave the country. A worker with specialized skills may have a strong reservation option because if they decline a job offer, they are likely to have another offer. But the employer may also have other qualified candidates. In the process of interviewing the worker and making an offer, what could the firm possibly do to increase their bargaining power?
 - Another case, an entrepreneur who wanted to sell their business - under what circumstances might they have structural power? What factors might affect their bargaining power?

B. Topic 2 - Angela and Bruno: Three Cases

Angela and Bruno

- We model an interaction between a farmer, Angela, who produces grain, and Bruno, who owns the land she farms.
- The amount of grain produced depends on how many hours Angela works each day.
- We can think of the grain they each obtain as their income from the interaction; it is Angela's only source of income, so if she receives too little grain she will starve.
- Depending on the rules of the game, and her alternative options, how much power Bruno has over Angela differs.

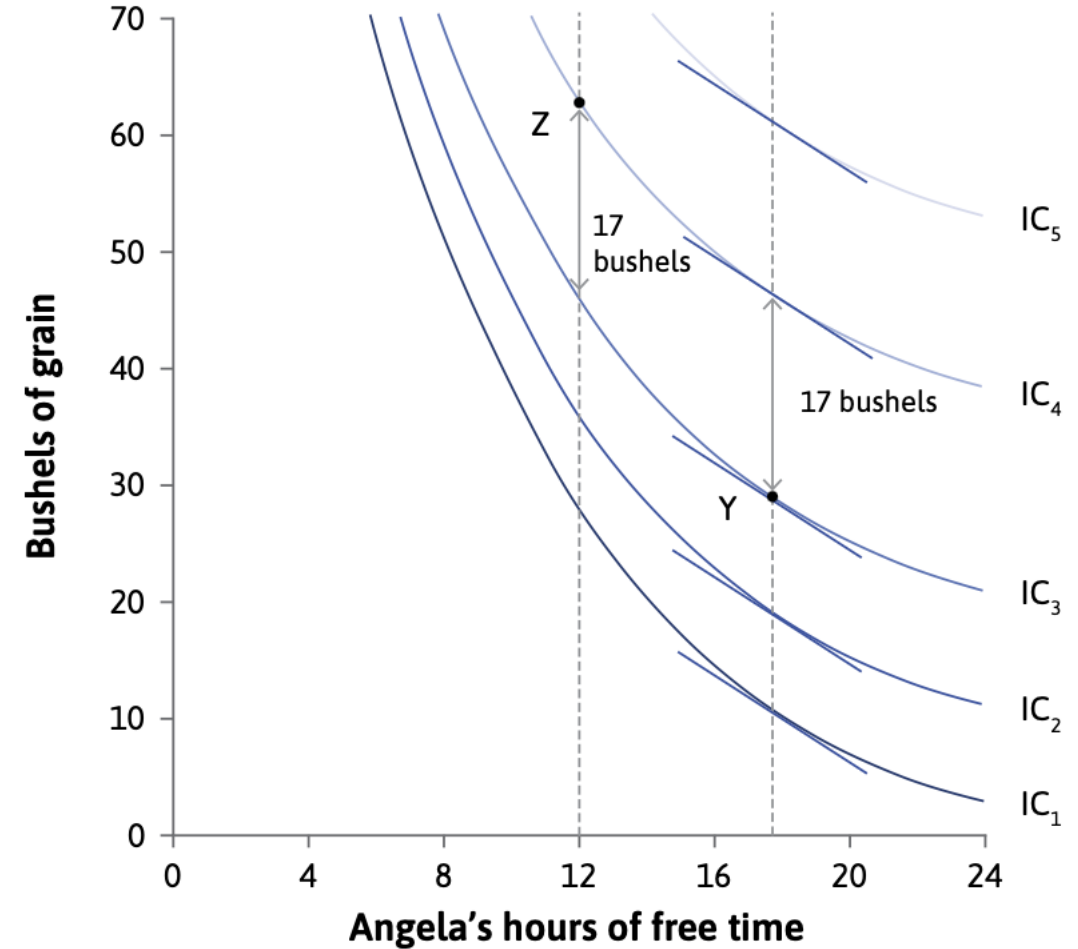
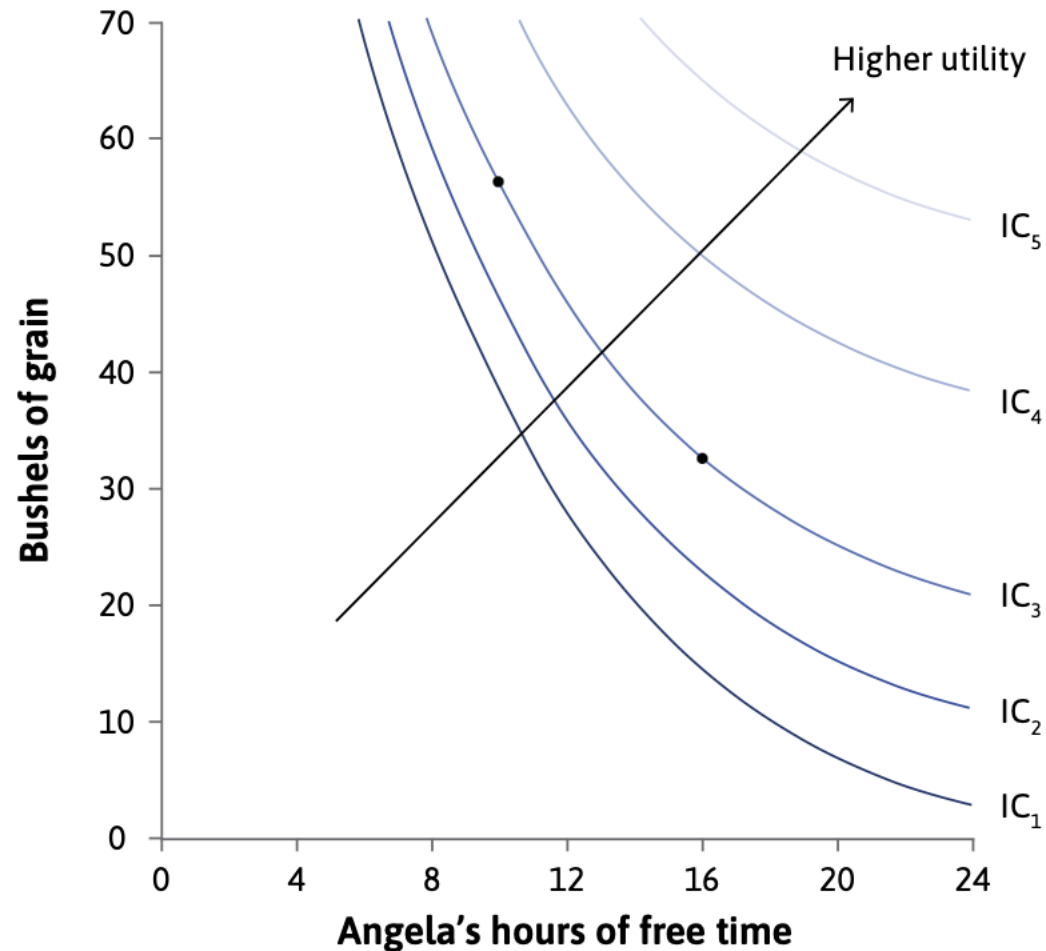
Angela and Bruno

While the institutions differ, the **preferences** of Bruno and Angela, and the **technology** that Angela uses to produce grain, are the same in each case:

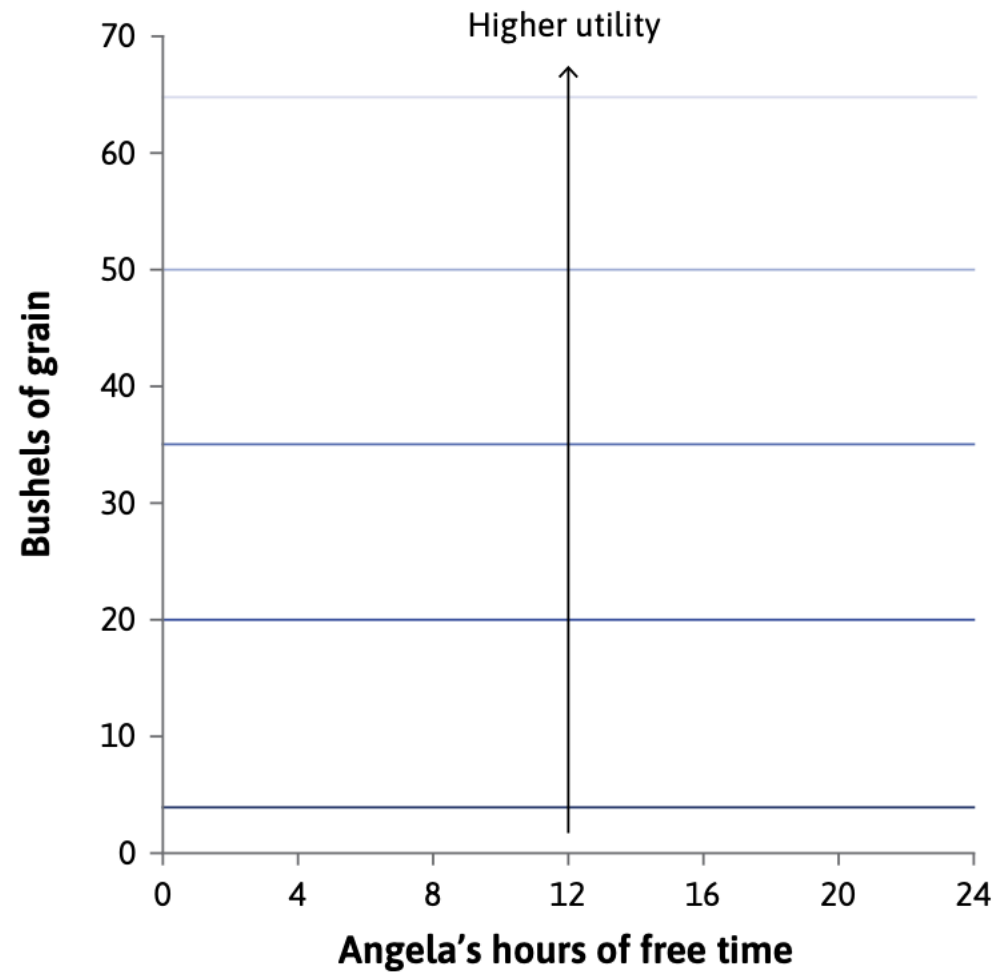
- Angela wants: The best-for-her feasible combination of grain and free time, according to her preferences.
- Bruno wants: As much grain as possible (he is not doing any work).
- *The **feasible set** of hours of Angela's work and the total amount of grain to be divided among the two, as given by the farming technology (the **production function**).*

Angela's preferences

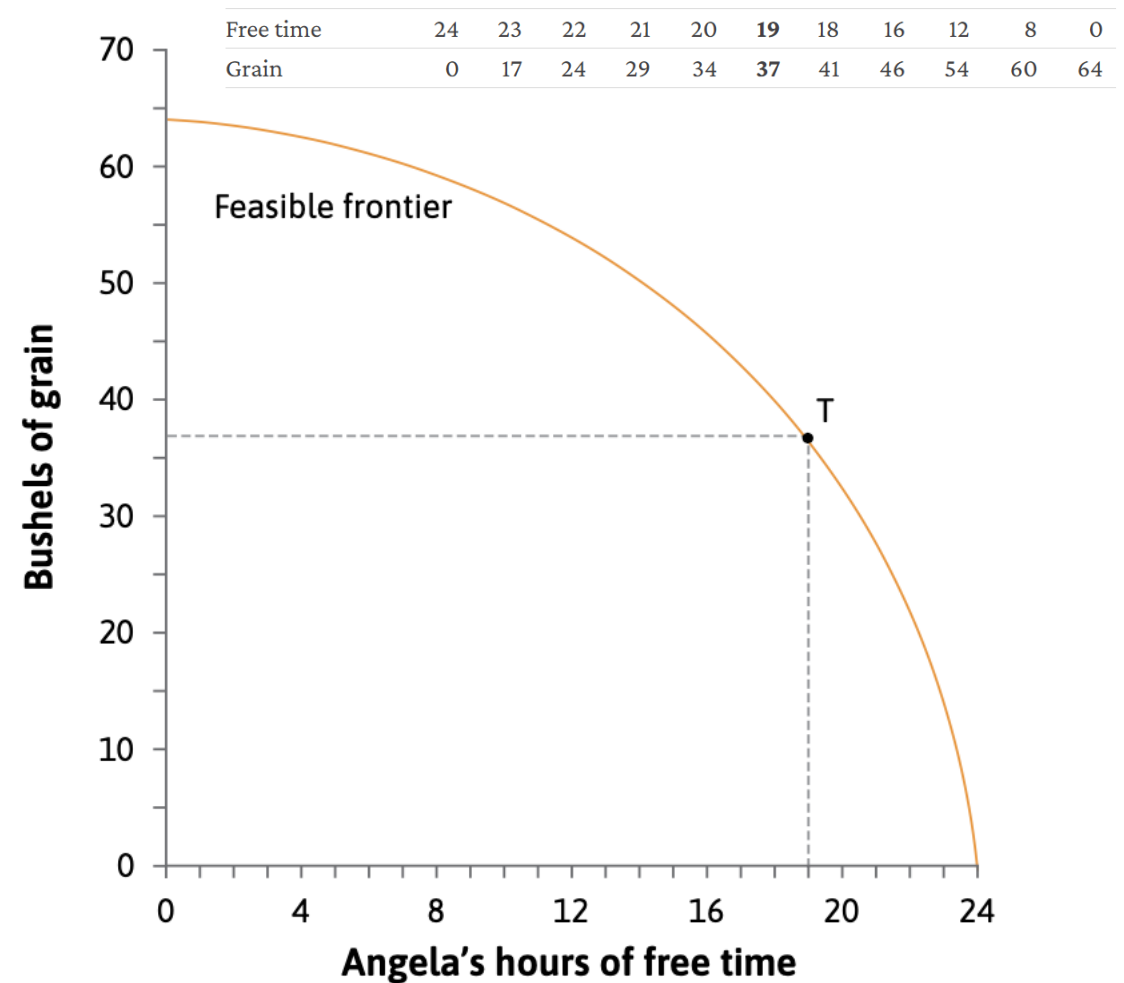
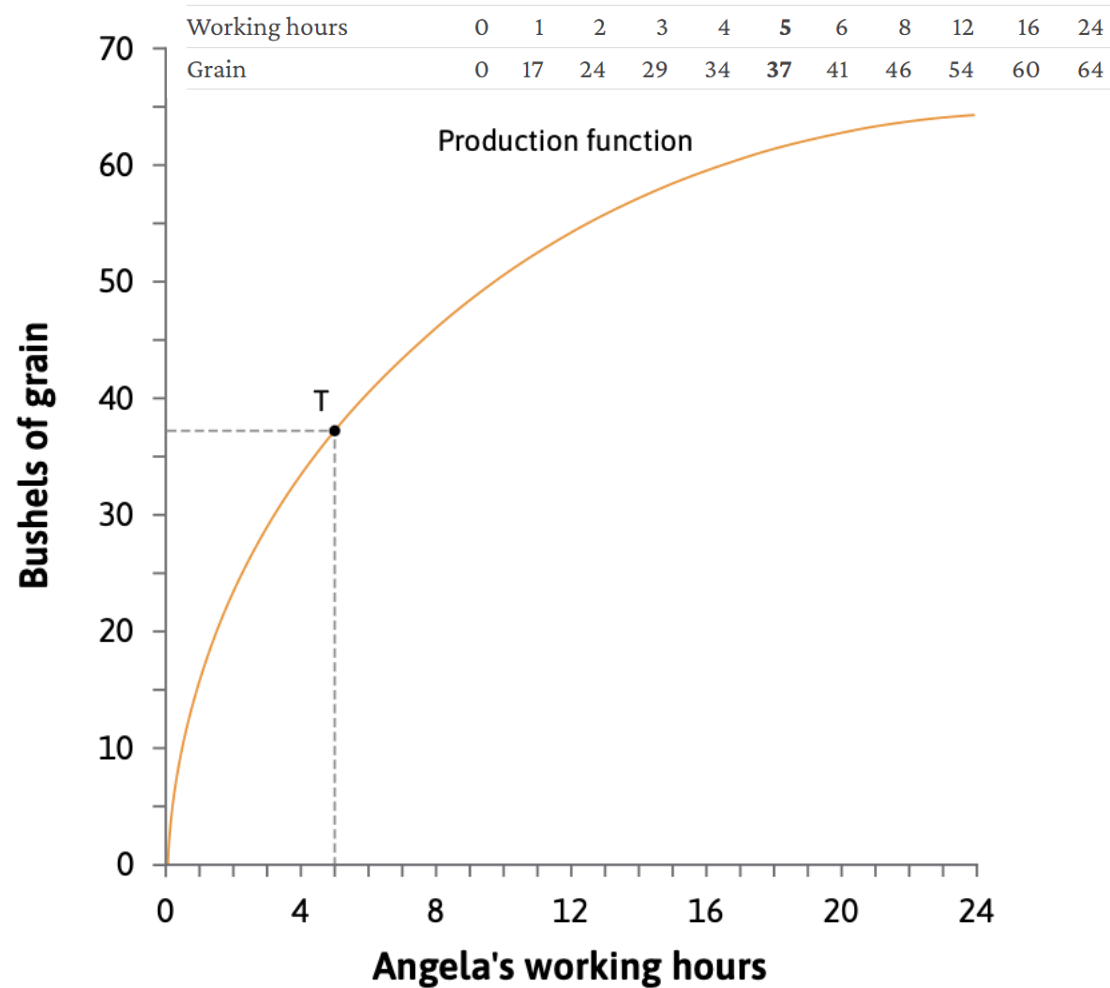
Angela's MRS depends on the amount of free time she has, but does not change if she receives more or less grain.



Bruno's preferences



Production function and feasible frontier



An independent farmer		
Baseline case: Angela owns the land		
Angela owns the land herself. The government protects her right to exclude others from the land (or its produce).		
Angela decides: how many hours to work and how much grain to produce and consume.		
A landowner and a farmer		
Bruno owns the land and Angela farms it		
What happens depends on the institutional setting		
Case 1	Case 2	Case 3
<i>Forced labour</i>	<i>A take-it-or-leave it contract</i>	<i>Bargaining in a democracy</i>
The rules of the game		
Bruno can force Angela to work for him producing grain which he owns.	Bruno offers Angela a contract (either an employment contract, or a farm tenancy) which she can accept or reject.	Bruno offers Angela a contract. Angela can accept the contract, reject the contract, or negotiate for alternative
Bruno decides: how many hours Angela must work, and how much grain she can consume.	There is no scope to negotiate over the contract terms. Bruno cannot physically threaten Angela, and she can refuse the offer and seek work elsewhere.	contract terms. Angela and other farmers are able to vote, enabling them to elect a government that will pass legislation limiting the
Angela decides: Obey, or attempt escape or, with other farmers, revolt (in both latter cases risking death).	If necessary, the government will protect Bruno's property rights and enforce a contract.	maximum number of hours worked and stipulating a minimum wage equal to what she would have earned in Case
	Bruno decides: what contract to offer Angela.	2.
	Angela decides: Accept or reject Bruno's contract. Also, if she accepts a tenancy, how many hours to work.	Bruno and Angela bargain: Angela, and others, vote in order to improve their options. Then she and Bruno negotiate a contract.

Figure 5.6 The rules of the game in different institutional settings.

Baseline case: Angela owns the land

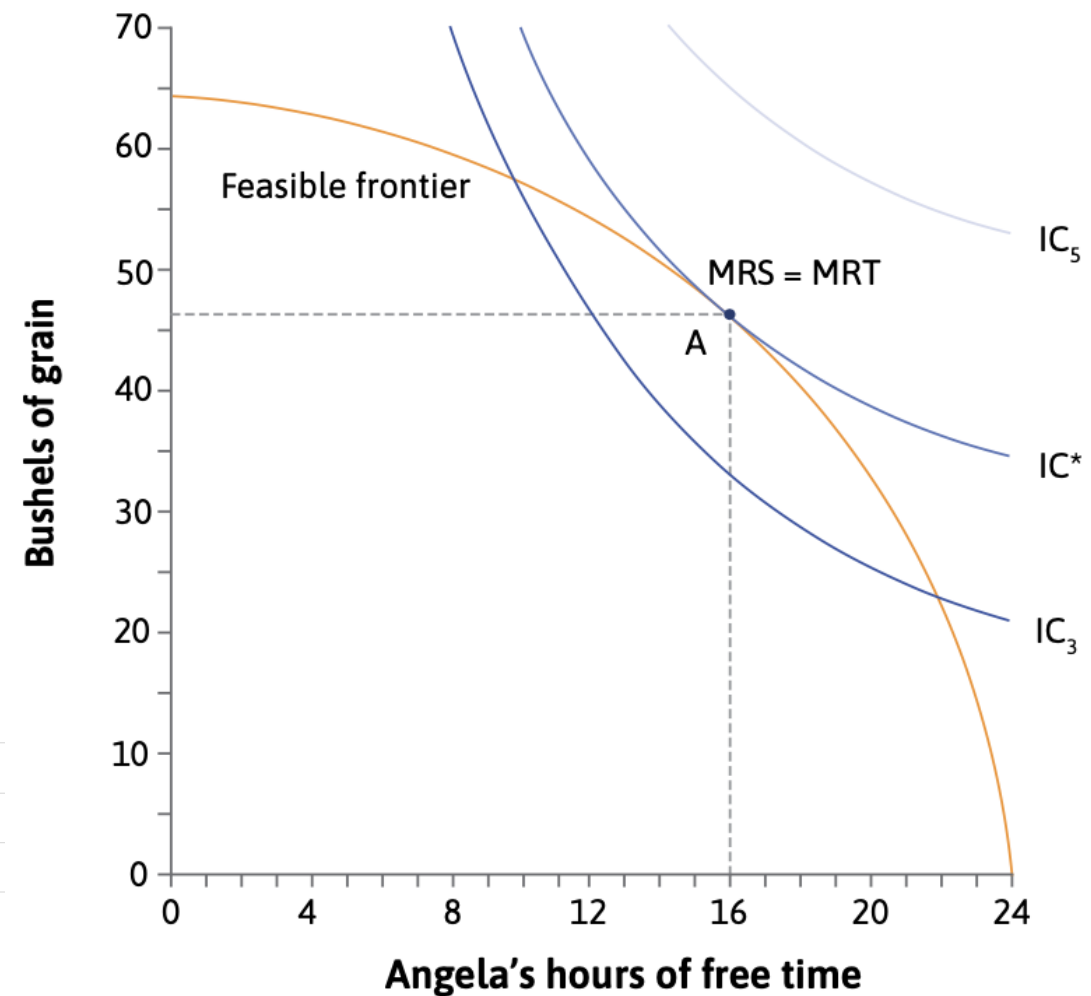
MRS = MRT for maximum utility

At point A, the indifference curve is tangent to the feasible frontier.

So her two trade-offs balance out: her marginal rate of substitution (MRS) between grain and free time (the slope of the indifference curve) is equal to the MRT (the slope of the feasible frontier).

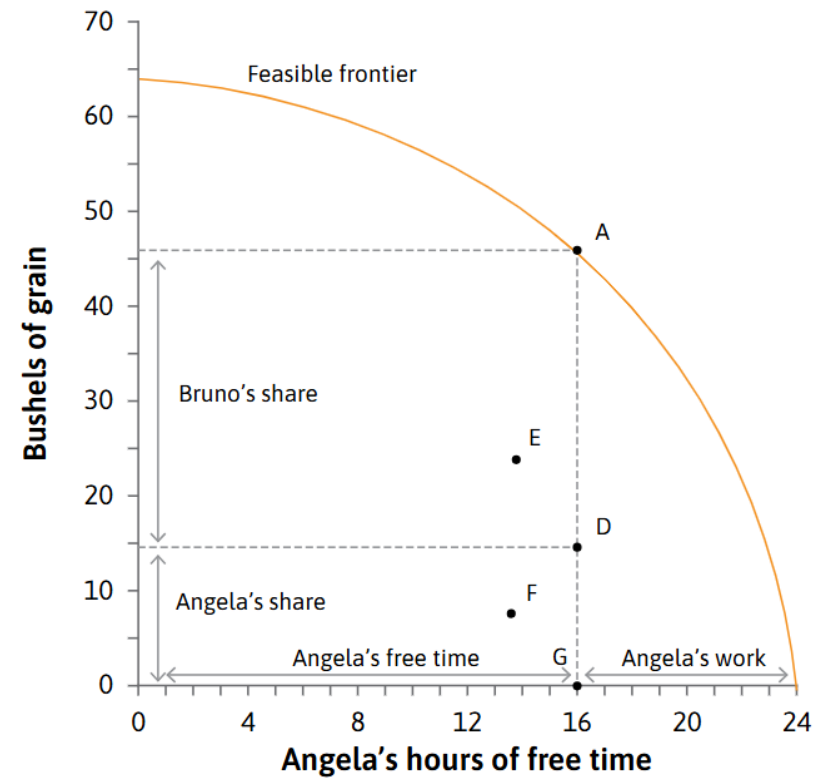
Angela's hours of free time	16
Angela's bushels of grain	46
Bruno's bushels of grain	n/a (Bruno is not a character in this scenario)

Figure 5.8 The outcome in the baseline case.



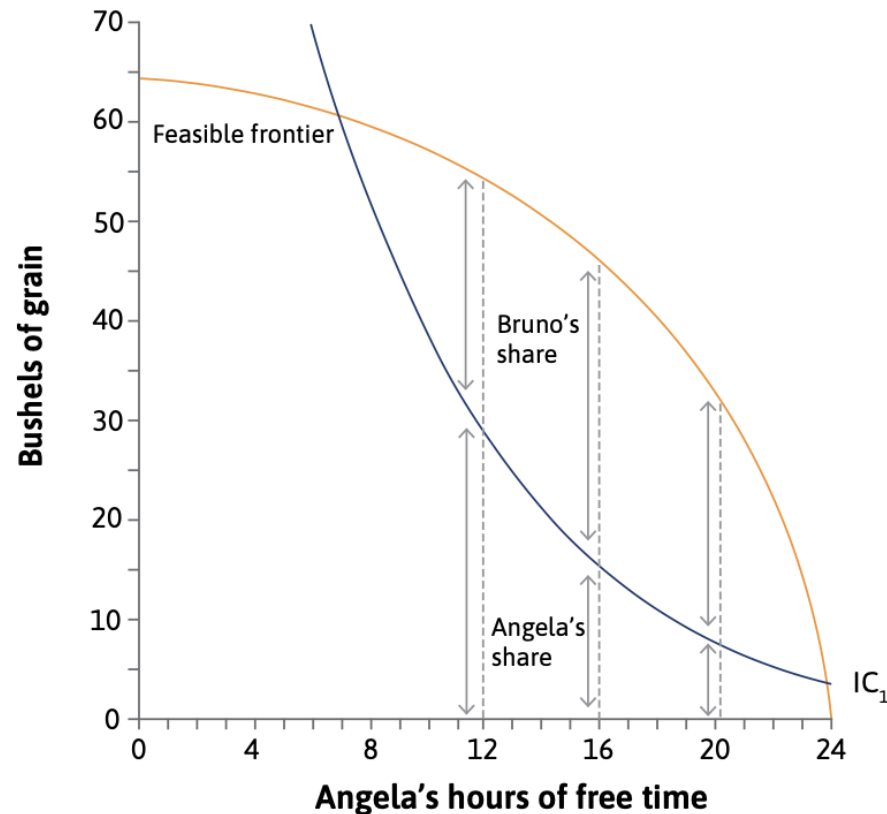
Case 1: Forced labour

Bruno owns the land. He forces Angela to work on his land producing grain, threatening her with physical harm if she does not comply with his orders.



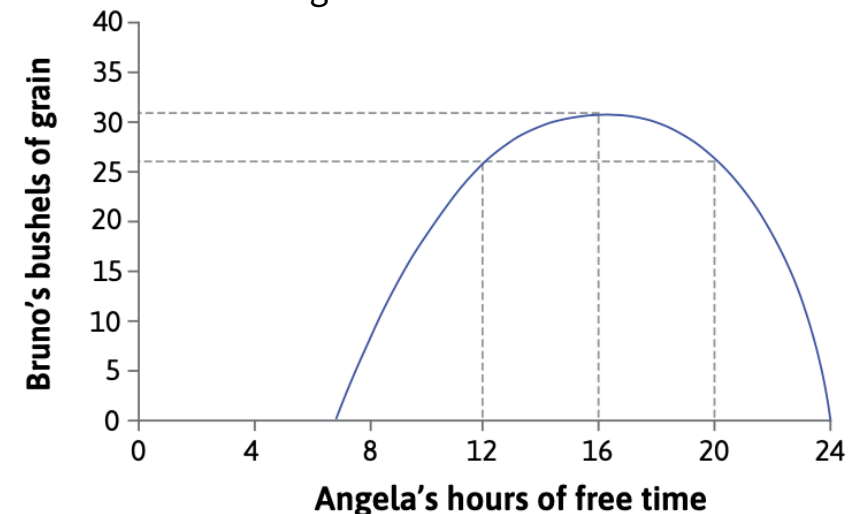
Case 1: Forced labour

Bruno owns the land. He forces Angela to work on his land producing grain, threatening her with physical harm if she does not comply with his orders.



Bruno wants to maximize his own amount of grain

- Bruno will get the most grain at allocation D, where he commands Angela to work eight hours, producing 46 bushels of grain (point A).
- At 16 hours of free time, $MRS = MRT$. Bruno would keep 31 bushels of grain, and give Angela the remaining 15 bushels.



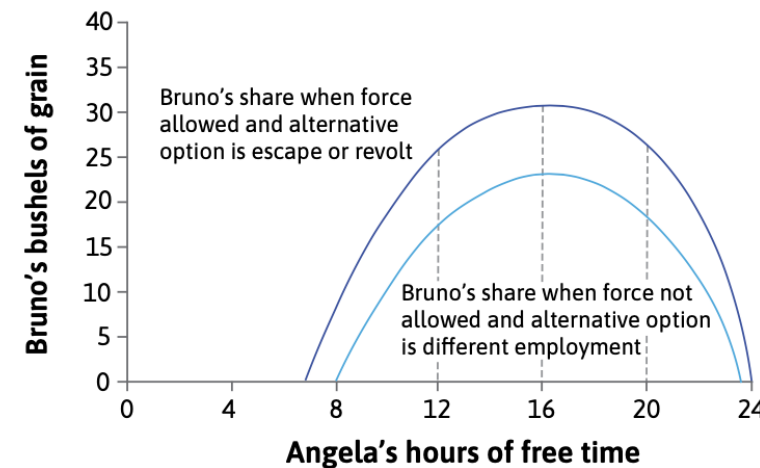
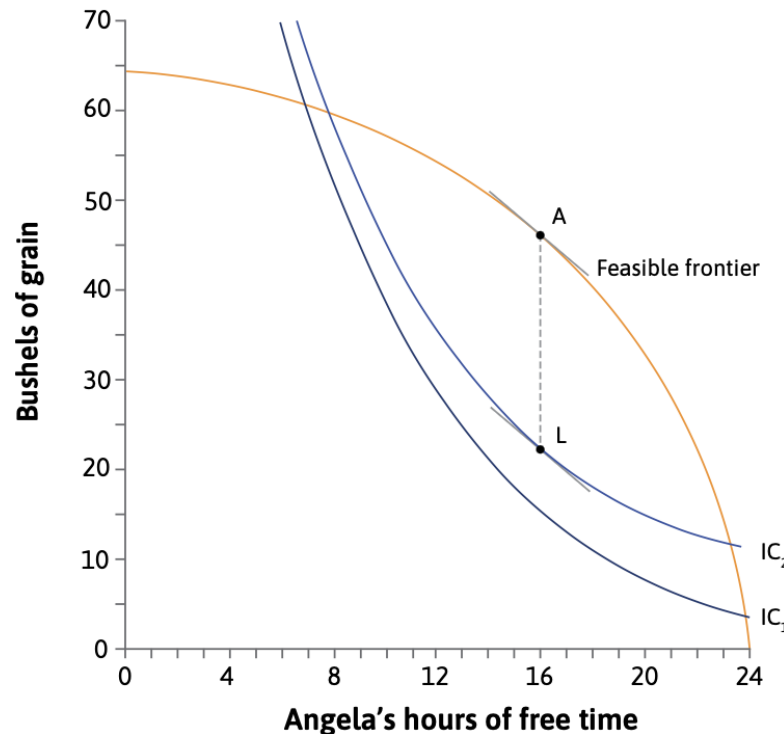
Case 1: Forced labour

Angela's hours of free time	16	
Angela's bushels of grain	15	
Bruno's bushels of grain	31	
Angela's economic rent (bushels)	0	She gets the same level of utility as in her best alternative option (disobeying)
Bruno's economic rent (bushels)	31	His best alternative is 0 (if Angela disobeys)

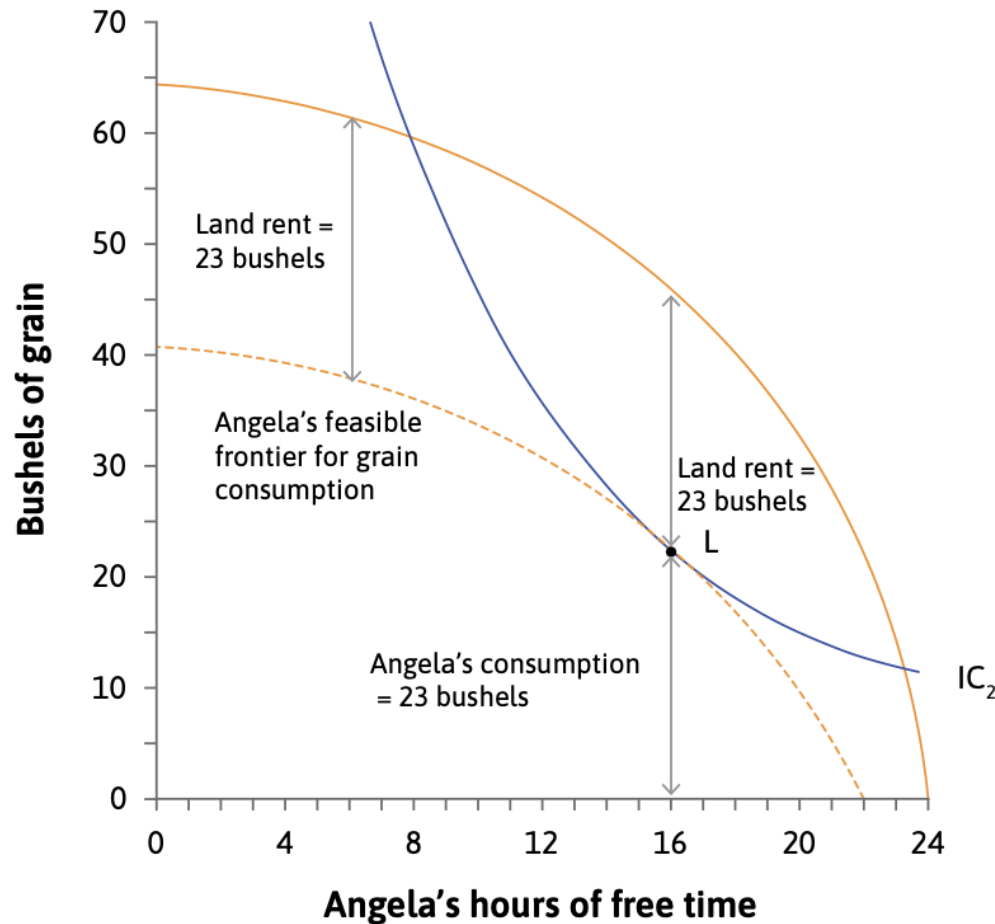
Figure 5.11 The outcome of Case 1.

Case 2: A take-it-or-leave-it employment contract

Bruno owns the land that Angela works. But the government protects Angela from being forced to work, as well as Bruno's property rights as a landowner. Furthermore, the legal system will enforce contracts between Bruno and Angela.



Case 2: A take-it-or-leave-it tenancy contract



Bruno offers Angela a tenancy: he specifies the rent that she must pay him to use the land and leaves it to her to decide how to use it.

- The best Angela can do if she has to pay a land rent of 23 bushels is to work for eight hours, and consume 23 bushels of grain (point L).
- This gives her the reservation level of utility, so she will be willing to accept Bruno's offer. (Any other choice of hours would give her less than reservation utility.)
- Bruno has achieved exactly the same outcome for himself, and Angela, as under the employment contract.

Case 2: A take-it-or-leave-it

Angela's hours of free time	16	
Angela's bushels of grain	23	
Bruno's bushels of grain	23	
Angela's economic rent (bushels)	0	She gets the same level of utility as in the best alternative of working elsewhere
Bruno's economic rent (bushels)	23	His best alternative is 0 (if she refuses his offer)

Figure 5.15 The outcome of Case 2.

Bargaining in a democracy

Bruno owns the farm and Angela is Bruno's employee. Bruno's property rights and Angela's right not to agree to a contract are protected by law. Angela is a voter and is free to seek change in the legal institutional arrangements, together with other farmworkers. She has democratic political rights.

Suppose that Angela, with others who work on neighbouring farms, lobbies the government to improve their conditions. They want working hours for farm labourers restricted to four and a half hours per day. They also want to get at least the same amount of grain as under the voluntary take-it-or-leave-it contract agreement which was 23 bushels of grain.

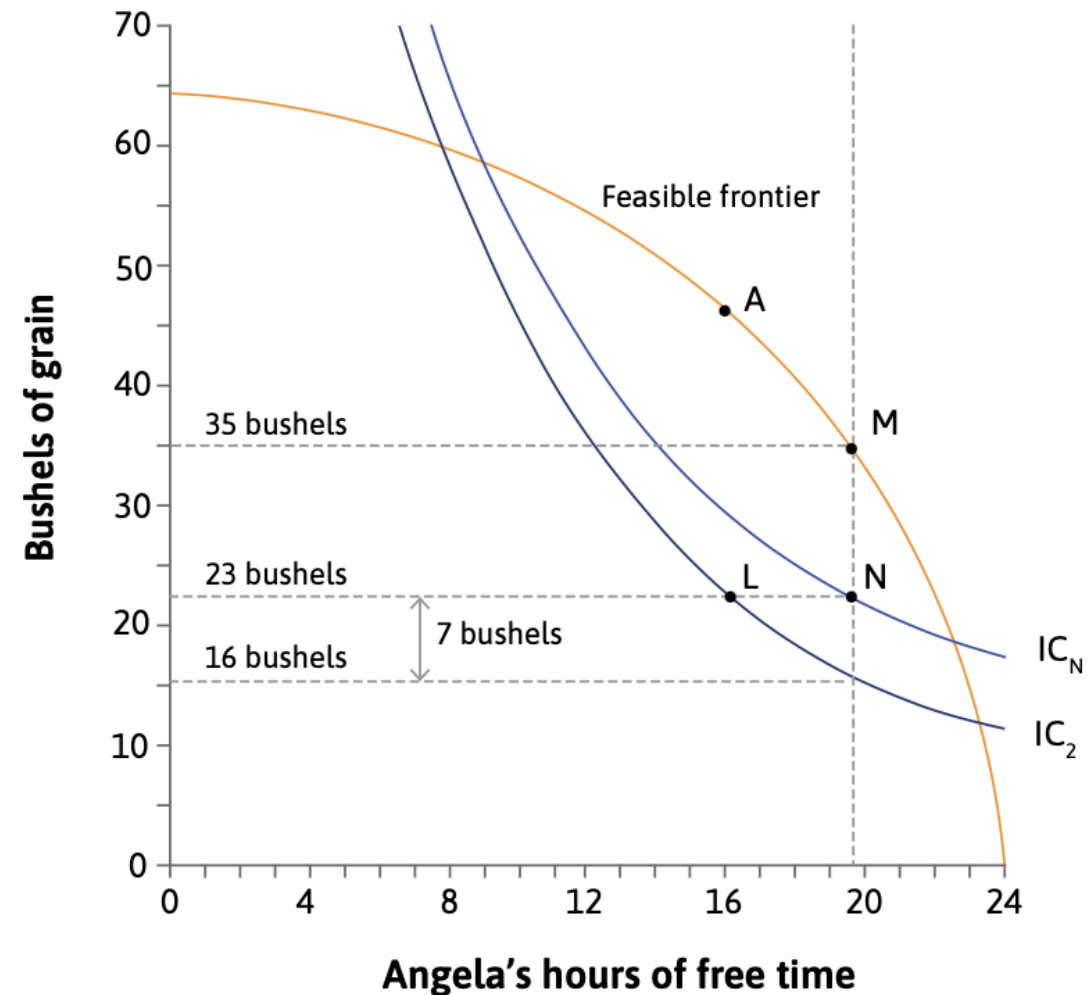
Case 3: Bargaining in a democracy

Angela, and other workers, will vote for the political party that agrees to implement their demands. They use their ability as voters to make legislative demands on the government, influencing their employment contracts.

The government agrees to the demands of the farm labourers, hoping to secure their vote in subsequent elections.

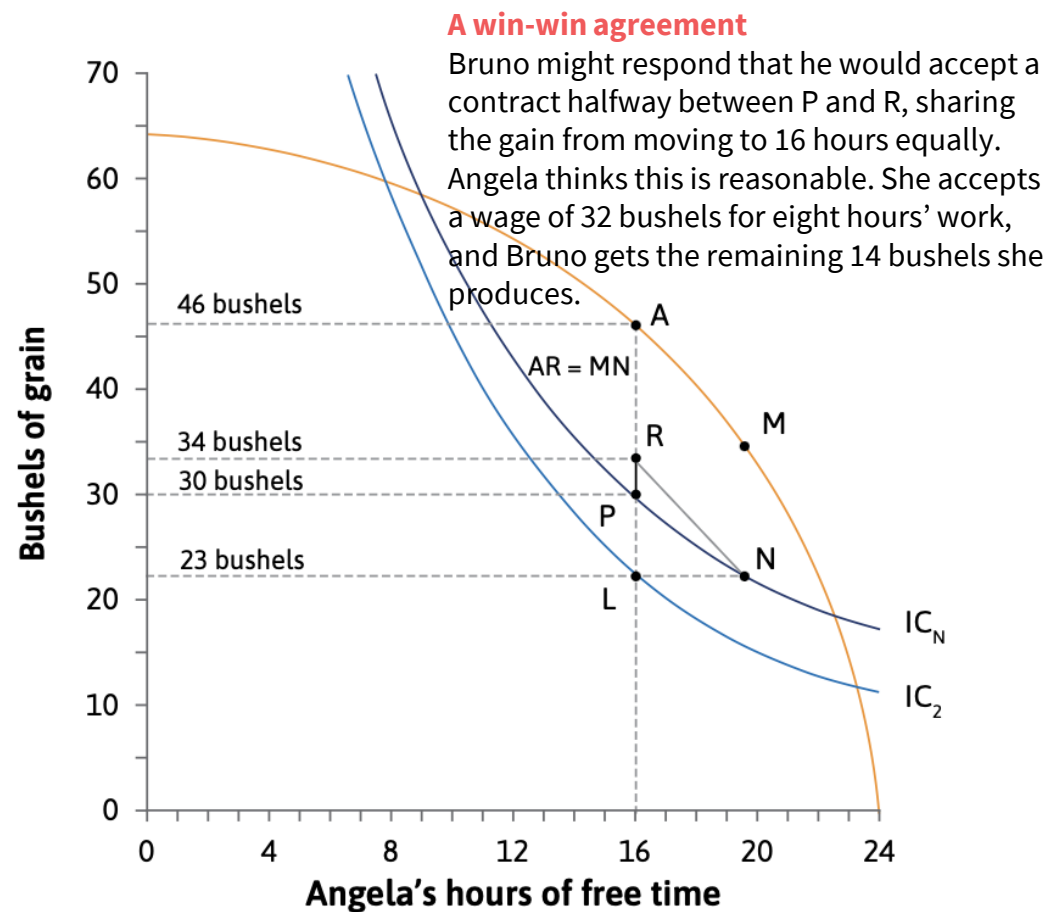
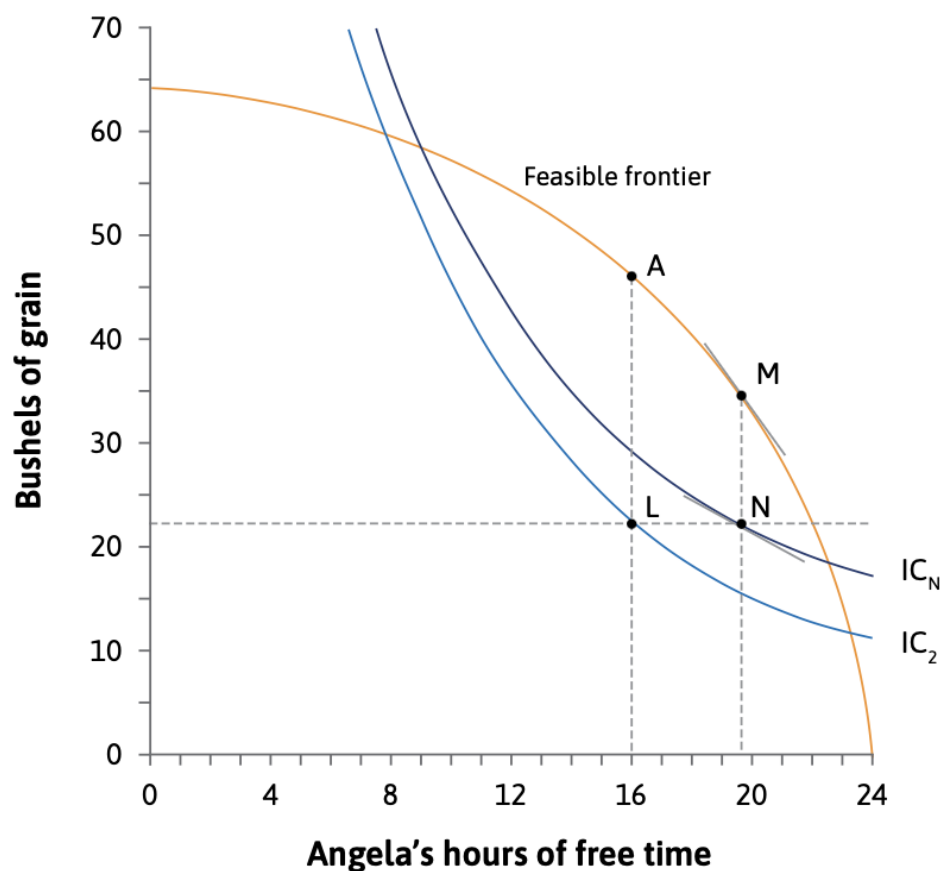
Angela will produce less grain and get higher utility

Under contract N, Angela gets the same wage as in contract L (Case 2), but more free time. This contract puts her on her new reservation indifference curve IC_N , seven bushels above IC_2 . Measured in terms of grain, her utility is seven bushels higher.



Pareto improvement through negotiation

We will suppose that the new law allows a longer workday if both parties voluntarily agree, provided that the fall-back position is a 4.5-hour day if no agreement is reached.

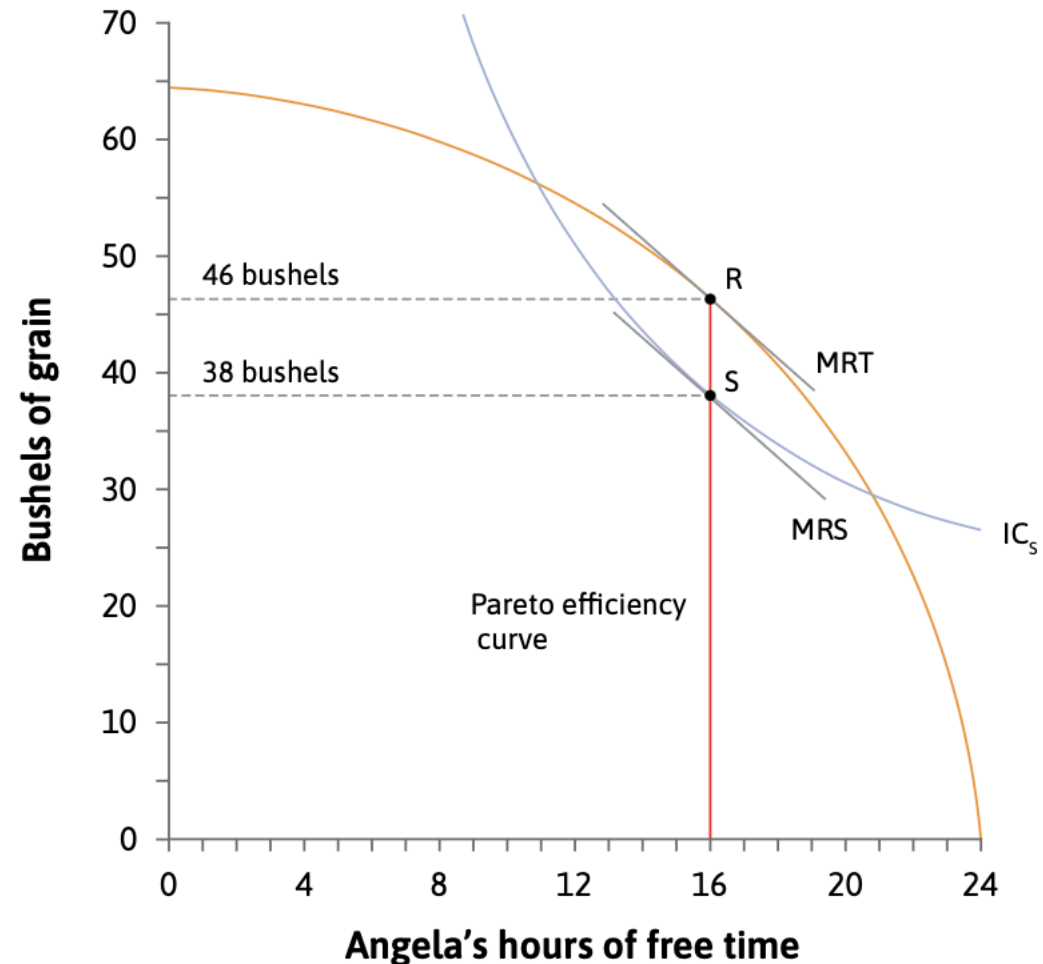


The Pareto efficiency curve

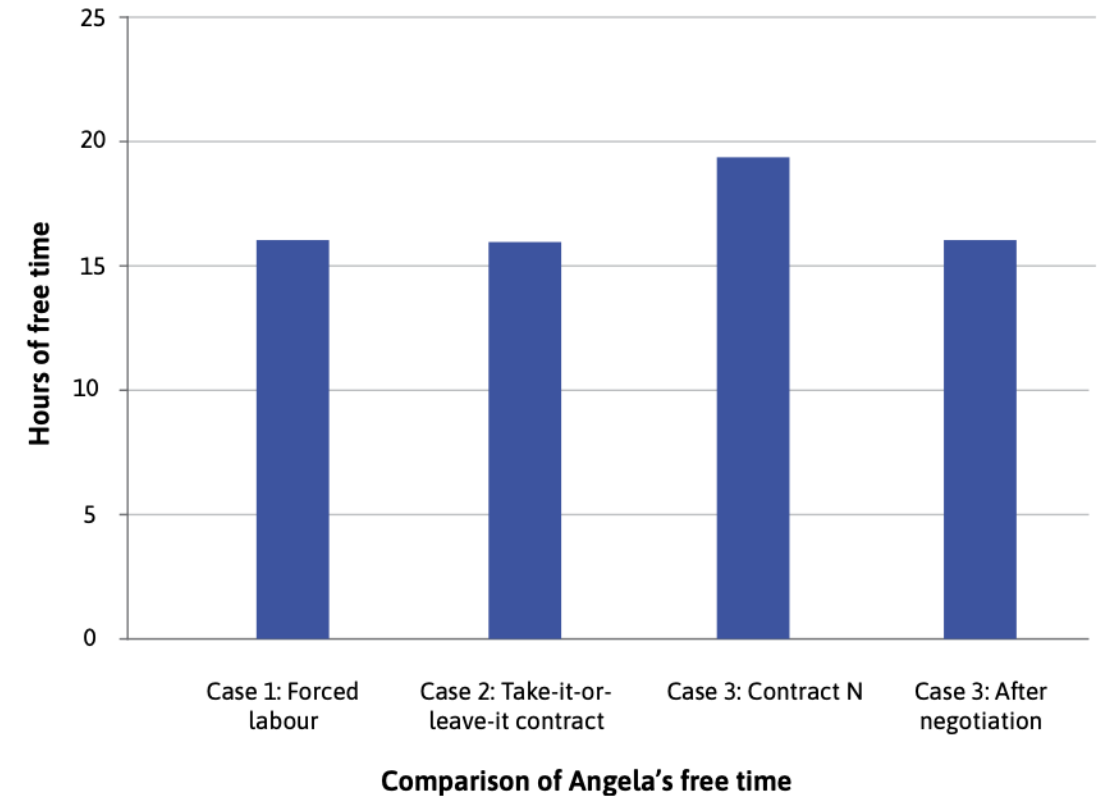
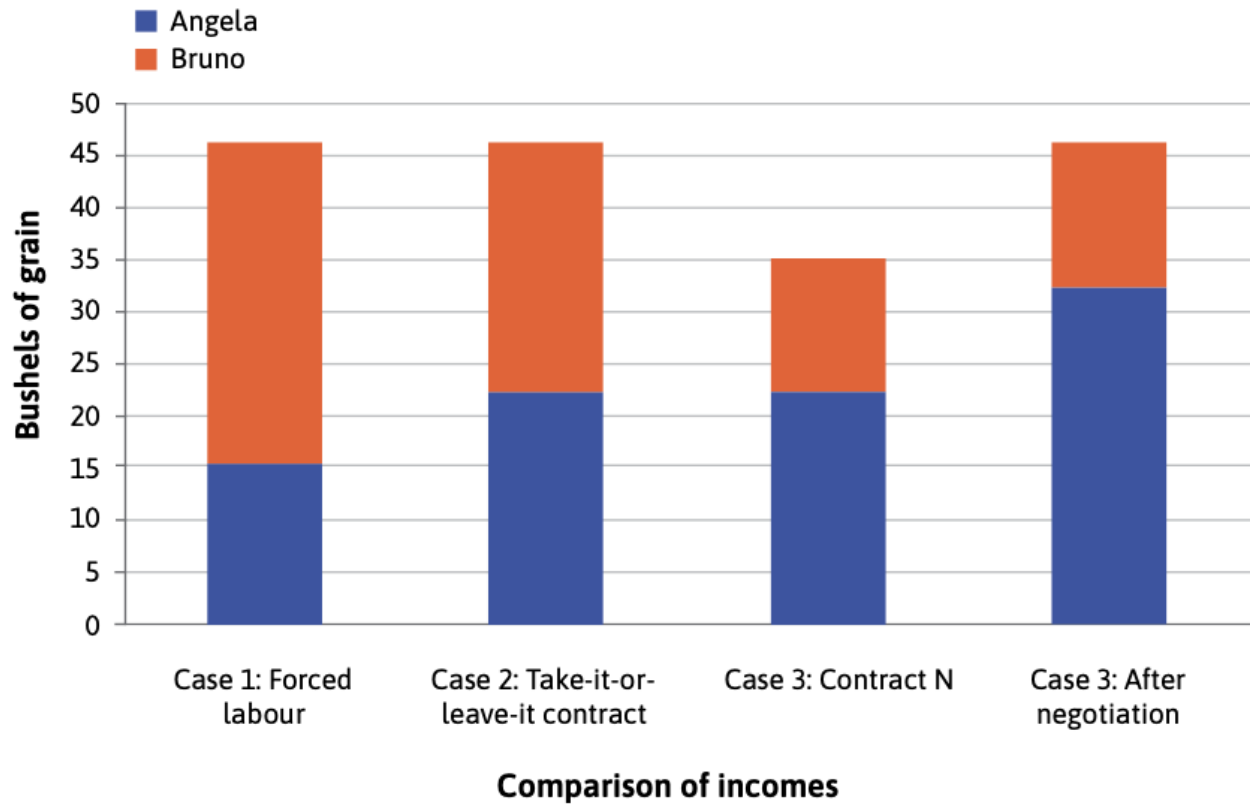
To be Pareto efficient, an allocation must have two important properties:

1. The MRT on the feasible frontier is equal to the MRS on Angela's indifference curve.
2. No grain is wasted.

The set of all Pareto efficient allocations is called the **Pareto efficiency curve**.



A summary of the different cases



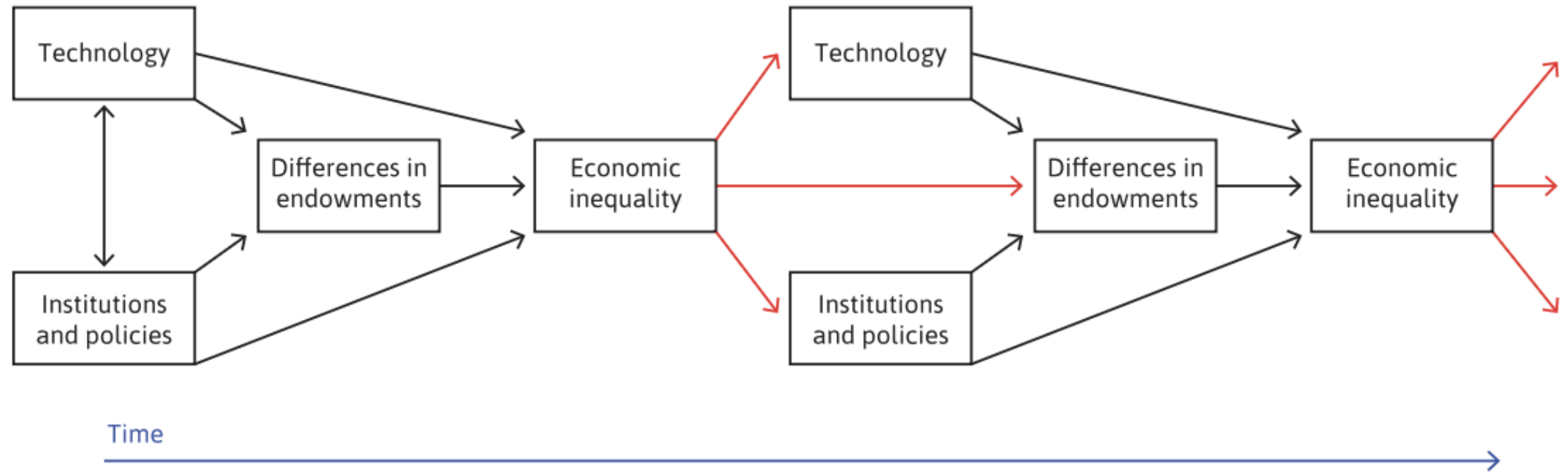
C. Topic 3 - Measuring, understanding and addressing inequality

Endowments

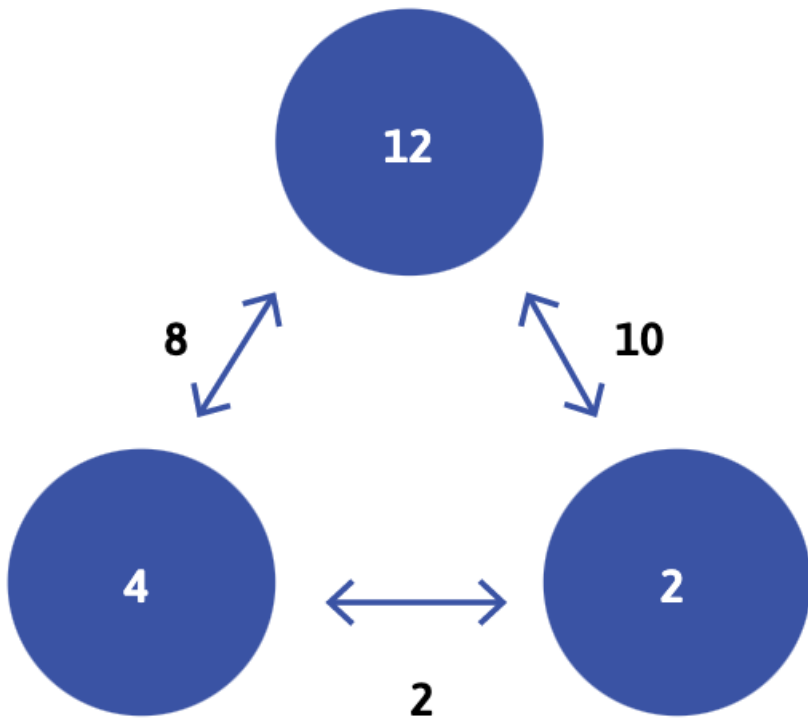
An individual's income depends on things that allow them to receive income. These **endowments** can be things they own, have, or are. They include:

- Financial wealth
- The physical assets they own
- Intellectual property such as copyrights or patents.
- Knowledge, skills and other personal attributes
- Their race, gender, age, and other aspects
- Their citizenship and whether they have a visa, which determine whether they can legally work in a particular country.

Economic inequality



Gini coefficient



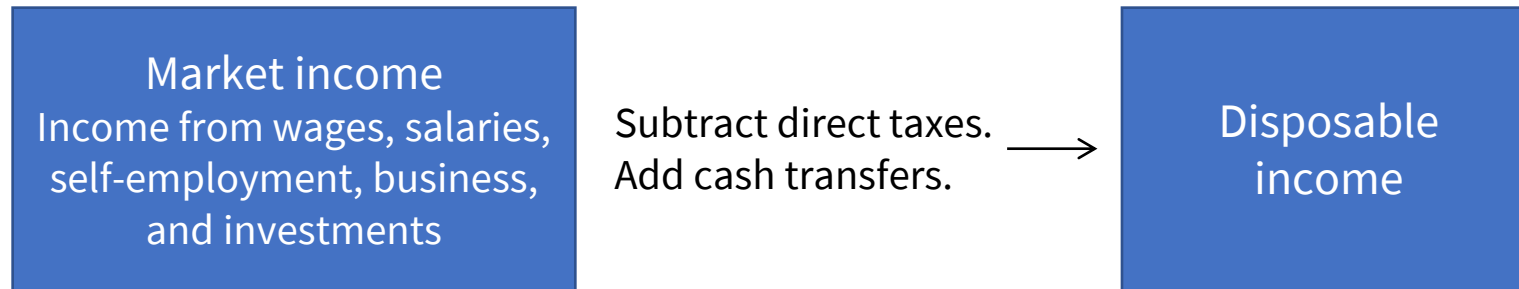
The Gini coefficient for income is equal to one-half times the average difference in income divided by the average income.

The average difference in income is $(10 + 8 + 2)/3 = 20/3 = 6.67$.

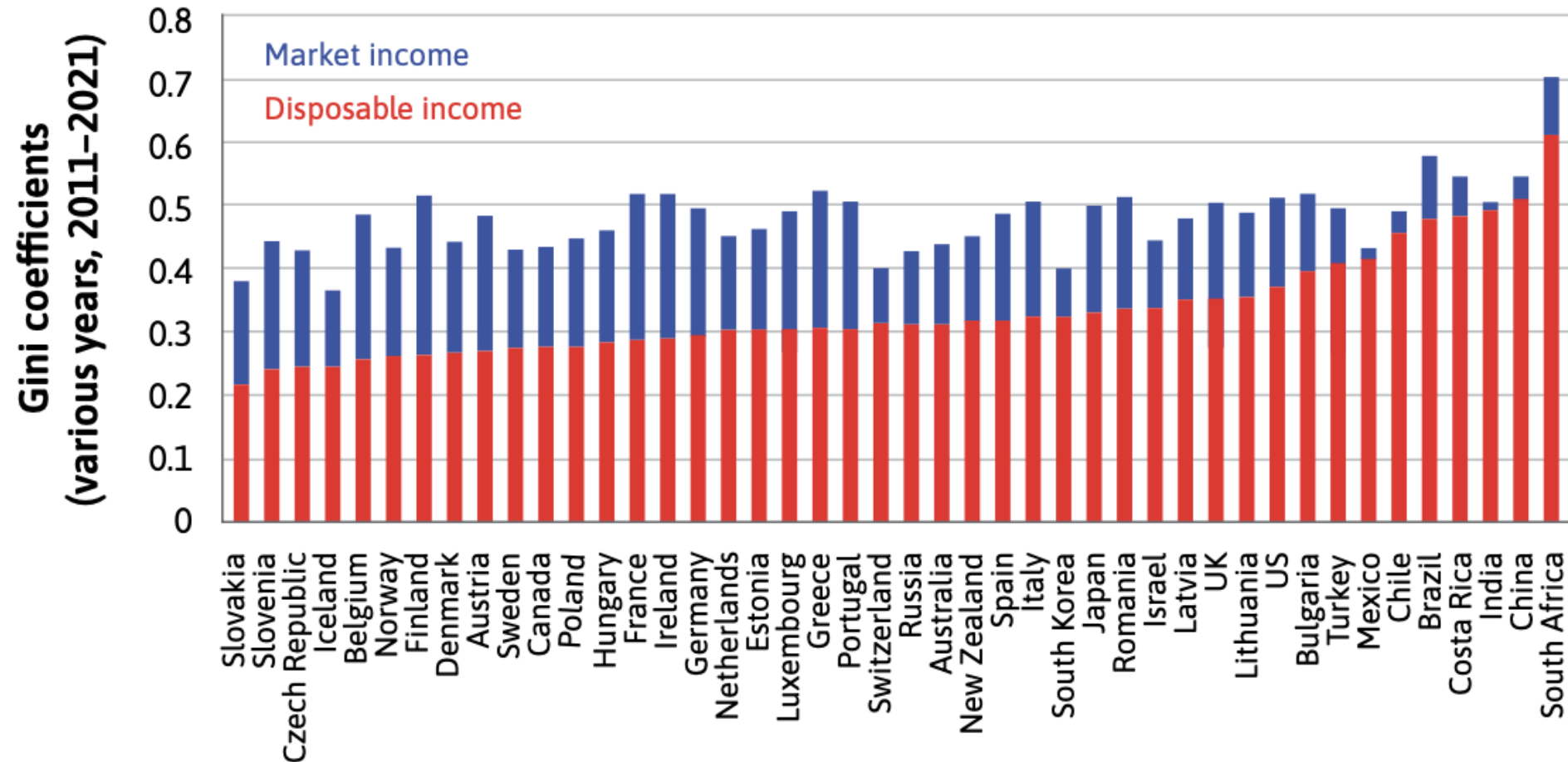
The average income is $(12 + 4 + 2)/3 = 6$.

Therefore, the Gini coefficient is $(0.5 \times 6.67)/6 = 0.56$.

Market vs disposable income



Market vs Disposable Income



In-class Activity (1)

- Choose one of the graphs in the article (see Cyber Campus Week 5) and describe, in your opinion, the most interesting feature.
- Explain what is meant by the Gini coefficient.
- Explain what the article suggests has happened to income inequality since the 1970s.
- Look up income inequality statistics for your country or a country of your choice (Google it) and compare it with those in the article.
- Write a brief report.

In-class Activity (2)

- Briefly explain the IMF findings from the article (see Cyber Campus Week 5) and say whether you intuitively agree or not, and why.
- Based on the information in the article and in the unit, to what extent is low post-tax inequality a result of government redistribution?
- Explain the challenges involved with designing a tax system that effectively reduces inequality.
- Act as policymakers tasked with reducing income inequality in a country of your choice. Propose some policies.
 - Ask AI for a feedback on your policies.
 - For example, “ChatGPT, I'm acting as the finance minister of a country facing high income inequality and slow economic growth. I propose increasing the top marginal tax rate and investing in education and healthcare. Can you simulate the potential impacts of these policies on the economy and society over the next five years?”
- Write a brief report.

Summary

1. Two criteria used to evaluate outcomes
 - Objective: **Pareto efficiency**
 - Subjective: **Fairness (substantive and procedural justice)**
2. **Allocations** depend on preferences and power/institutions
 - **Institutions** can determine **reservation options** and **bargaining power**.
 - The **Gini coefficient** measures the inequality of allocations.
 - Public policies can make allocations more efficient or fairer.

In the next unit

- The role of other institutions: **firms** and **markets**
- How firms work: owners, managers, and employees
- Why unemployment is a feature of the way labour markets function